

Question:

There exist real numbers x and y , both greater than 1, such that $\log_x(y^x)=\log_y(x^{4y})=10$. Find xy .

Reasoning + Answer:

<think>

1. **Understand the given problem :**

We are given two logarithmic equations with real numbers (x) and (y) both greater than 1 :

$$\log_x(y^x) = 10 \quad \text{and} \quad \log_y(x^{4y}) = 10$$

2. **Rewrite the given equations using logarithmic properties :**

- For the first equation :

$$\log_x(y^x) = 10$$

Using the logarithm power rule $(\log_b(a^c) = c \log_b(a))$, we get :

$$x \log_x(y) = 10$$

- For the second equation :

$$\log_y(x^{4y}) = 10$$

Similarly , using the logarithm power rule :

$$\log_y(x^{4y}) = 4y \log_y(x) = 10$$

3. **Set up the equations to solve for $(y \log_x(y))$:**

From the first equation :

$$x \log_x(y) = 10 \implies \log_x(y) = \frac{10}{x}$$

From the second equation :

$$4y \log_y(x) = 10 \implies \log_y(x) = \frac{10}{4y} = \frac{5}{2y}$$

4. **Express $(\log_x(y))$ and $(\log_y(x))$ in terms of each other :**

Recall the change of base formula for logarithms :

$$\log_x(y) = \frac{1}{\log_y(x)}$$

Therefore :

$$\log_x(y) = \frac{1}{\log_y(x)} \implies \frac{10}{x} = \frac{1}{\frac{5}{2y}} \implies \frac{10}{x} = \frac{2y}{5}$$

5. **Solve for (xy) :**

From :

$$\frac{10}{x} = \frac{2y}{5}$$

Cross-multiplying gives :

$$10 \cdot 5 = 2y \cdot x \implies 50 = 2xy \implies xy = 25$$

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<answer>

25

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